

Q1. What is meant by negative resistance?

Tarun Sharma

Ajmer

A1. The resistance of a device is defined as the ratio of voltage present across it to the current passing through it. This is denoted by $R=V/I$, where 'R' is the static resistance. It can also be defined as $r=dV/dI$, where 'r' is

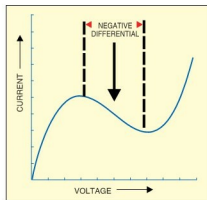


Fig. 1: I-V curve for a tunnel diode showing the negative resistance region



Tunnel diode

the dynamic or differential resistance.

For most materials, the current is proportional to the applied voltage and these two ways of specifying resistance are indistinguishable. Such materials are said to obey Ohm's Law. Some materials, under certain conditions, do not follow Ohm's Law.

A negative resistor is theoretical

and it does not exist as a discrete component. Negative resistance is a property of some electric circuits, where an increase in the current through a port results in decreased voltage across that port.

Some types of diodes such as tunnel diodes and Gunn diode exhibit negative resistance in some part of their operating range. Differential negative resistance of a tunnel diode is illustrated in Fig. 1. Devices and systems exploiting this effect are widely used to make amplifiers and oscillators in the microwave, millimetre wave and terahertz frequency ranges.

Q2. How is a neutral line obtained at a power station?

Prakash

Ahmedabad

A2. Neutral is not obtained at a power station. It is a part of the transmission bus, which has three lines for three phases and one line for neutral. It is a 3-phase, 4-wire system.

In a 3-phase low-voltage distribution system, the preferred installation should consist of a five-wire system. That five-wire system would consist of 3-phase conductors, a neutral conductor and a separate ground conductor. In normal practice, the ground conductor is often the building's ground. Four-wire circuits offer flexibility, since a load may be connected line-to-line (as in 3-phase motors) or line-to-neutral.

Transmission lines often feature a ground. But this is solely for fault and

lightning protection and is not connected to deliver electrical power.

Q3. What is the significance of 3dB gain and 3dB bandwidth?

Si Komal

Hyderabad

A3. First, let us define the terms dB, gain and bandwidth.

Decibel or dB is a logarithmic unit of measure of the ratio between two numbers. It is simply the ratio of output to input expressed in logarithmic form. The base of the log is 10.

The gain of an amplifier is the ratio of output to input power or amplitude. When measured in decibels, it is logarithmically related to the power ratio.

The bandwidth of an amplifier is usually defined as the difference between the lower- and upper-half power points. Half power points are

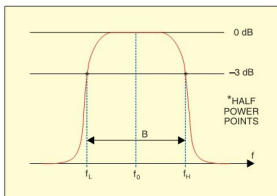


Fig. 2: 3dB bandwidth plot

the points on a gain-frequency plot where power has dropped by half or output voltage has dropped by $1/\sqrt{2}$ or 0.707. An amplifier will have two half power points, whilst a low-pass filter amplifier will have only one, the upper-half power point. A high-pass filter will have only the lower-half

KElement
LED TV

LCD 3D TV

Pocket Smart TV

Models:
Taupe
Woodbine
Purple
orchid
Endive
Bisque
Lilac

Learning Solutions for 21st Century
Price Starts from Rs. 4999

ktab
www.ktab.in
admin@ktab.in

Make your old TV to Internet TV

www.kelement.in
sales@kelement.in

{imagine innovate inspire}

Trade Enquiry
099943 55780
080560 45640
1800 425 3099